

SelfCheckGPT Reproduction and Extension: Cross-Lingual Empirical Research and Multi-Dimensional Feature Fusion for LLM Hallucination Detection

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Abstract

This study reproduces and enhances SelfCheckGPT, a zero-resource black-box hallucination detection framework for LLMs. We first replicated all five SelfCheckGPT variants (BERTScore, NLI, MQAG, Ngram, Prompt) on the WikiBio dataset, achieving results highly consistent with the original paper, with NLI and MQAG outperforming the baseline.

To further improve performance, we proposed an ensemble strategy fusing NLI and Prompt via grid search weighting, yielding a NonFact AUC-PR higher than the top single model.

We then extend the research to cross-lingual scenarios, constructing a Chinese-WikiBio dataset and testing adapted versions of Ngram and Prompt. While Prompt retains robustness, n-gram fails drastically due to Chinese linguistic characteristics. Attempts to improve via semantic/symbolic/reasoning/translation enhancements prove ineffective, revealing key challenges like the "Entity Paradox" and "hallucination propagation" in Chain-of-Thought (CoT) prompting.

Finally, we extended the scope to grey-box scenarios (common in local deployment) to investigate whether internal uncertainty metrics can replace computationally expensive sampling. We proposed the Learning-to-Check (L2C) framework, fusing internal uncertainty with external consistency via Random Forest. Our results demonstrate that internal metrics alone fail due to severe model miscalibration, confirming that SelfCheckGPT remains the dominant indicator even when internal states are accessible. L2C achieves a higher NonFact AUC-PR than all single models on Chinese data, validating that non-linear fusion of complementary features mitigates individual limitations.

1 Introduction

With the development of LLMs, It has been widely observed that models can confidently generate fictitious information. However, previous hallucination detection methods suffer from notable limitations: grey-box approaches rely on internal LLM output distributions, making them incompatible with black-box systems; methods dependent on external databases are complex, constrained by database coverage, and incur high reasoning costs; most lack generalizability, focusing solely on specific tasks rather than general LLM responses.

In response to these challenges, the EMNLP 2023 paper "Self-CheckGPT: Zero-Resource Black-Box Hallucination Detection for Large Language Models" [1] proposes SelfCheckGPT as a solution. It is zero-resource and black-box compatible, requiring no external databases or internal model information and operating solely through sampling and comparing multiple LLM responses to the same query. And it is tailored for hallucination detection in general LLM responses. Since the data in the paper demonstrates that SelfCheckGPT exhibits excellent performance, we aim to replicate it.

While demonstrating impressive performance in detecting non-factual content, SelfCheckGPT has limitations in cross-

lingual applicability and some other aspects. So after reproducing the SelfCheckGPT experiment, we conducted enhancement experiments and extended research. Notably, considering the growing trend of local LLM deployment where internal states (logits) are accessible, we investigated whether low-cost internal uncertainty metrics could replace expensive sampling-based checks. Our Learning-to-Check (L2C) experiment ultimately proves the irreplaceable value of external consistency checking even in grey-box settings.

2 Approach

2.1 Core Principle & Framework of SelfCheckGPT

SelfCheckGPT's core idea is based on the premise that factual information remains consistent across multiple stochastic samples of LLM responses to the same query, while hallucinated content tends to diverge or contradict across samples. It involves sampling multiple LLM responses to the same query and comparing their consistency: consistent responses indicate high factuality, while inconsistencies suggest potential hallucinations.

The framework operates as follows:

1. For a given user query, generate a target response R and N additional stochastic samples $\{S^1, S^2, \dots, S^N\}$ (default $N = 20$) using the same LLM.
2. For each sentence r_i in R , measure its consistency with the N samples using one of five consistency metrics.
3. Map consistency to a hallucination score $\mathcal{S}(i) \in [0.0, 1.0]$: scores close to 0 indicate high factuality, while scores close to 1 indicate likely hallucination.

2.2 Five Variants of SelfCheckGPT

There are five consistency detection methods adopted in this process: BERTScore (semantic similarity), NLI (natural language inference), MQAG (question-answering consistency), Ngram (language model probability), and Prompt (LLM self-evaluation). They cover levels from surface similarity to semantic reasoning.

2.2.1 SelfCheckGPT with BERTScore

Assess semantic similarity between the target sentence r_i and the most similar sentence in each sample. Lower similarity implies higher hallucination risk.

$$\mathcal{S}_{\text{BERT}}(i) = 1 - \frac{1}{N} \sum_{n=1}^N \max_k (\mathcal{B}(r_i, s_k^n))$$

where $\mathcal{B}(\cdot, \cdot)$ is the BERTScore between two sentences, and s_k^n is the k -th sentence in sample S^n .

In this work, RoBERTa-Large is used as the backbone model of BERTScore.

2.2.2 SelfCheckGPT with NLI

Use Natural Language Inference (NLI) to judge if r_i contradicts each sample S^n . Higher contradiction probability implies higher hallucination.

For SelfCheck-NLI, we use DeBERTa-v3-large fine-tuned to MNLI as the NLI model. The input for NLI classifiers is the sampled passage S^n concatenated to the sentence to be assessed r_i .

$$P(\text{contradict}|r_i, S^n) = \frac{\exp(z_c)}{\exp(z_e) + \exp(z_c)}$$

where z_e (entailment) and z_c (contradiction) are logits from the NLI model. The neutral class is ignored to ensure that the probability is bounded between 0.0 and 1.0. The SelfCheckGPT with NLI score for each sample S^n is defined as:

$$S_{\text{NLI}}(i) = \frac{1}{N} \sum_{n=1}^N P(\text{contradict}|r_i, S^n)$$

2.2.3 SelfCheckGPT with MQAG

We use the automatic multiple-choice question answering generation (MQAG) framework to measure consistency for SelfCheckGPT. We generate multiple-choice questions from r_i , answer them using R and each S^n , and measure answer consistency. Larger inconsistencies indicate higher hallucination. The key steps are as follows:

1. Generate 5 QA pairs (with 3 distractors each) from r_i via two generators (G1: question+answer; G2: distractors).
2. Filter unanswerable questions using an answerability score α .
3. Compute effective match/mismatch counts (N'_m, N'_n) via soft-counting (incorporating α).
4. Final score (Bayesian inference):

$$S_{\text{QA}}(i) = \frac{\gamma_2^{N'_n}}{\gamma_1^{N'_m} + \gamma_2^{N'_n}}$$

(γ_1, γ_2 are hyperparameters set to 0.8 for both β_1, β_2).

The models used are T5-Large (question/distractor generation) and Longformer-Large (answering/answerability).

2.2.4 SelfCheckGPT with Ngram

It has been found that 1-gram outperforms longer n-grams (2-5 grams) for hallucination detection. So we train a unigram language model on R and N samples. Hallucinated sentences have lower probability in this model.

Use the maximum negative log-probability of tokens in r_i (more sensitive to outliers than average):

$$S_{\text{n-gram}}^{\text{Max}}(i) = \max_j (-\log \tilde{p}_{ij})$$

where $\tilde{p}_{ij}$ is the token probability from the unigram model.

2.2.5 SelfCheckGPT with Prompt

Query an LLM to directly assess if S^n supports r_i , using a simple yes/no prompt.

The prompt template is as follows:

Context: $\{S^n\}$

Sentence: $\{r_i\}$

Is the sentence supported by the context above?

Answer Yes or No:

Score mapping: Yes $\rightarrow$ 0.0, No $\rightarrow$ 1.0, N/A $\rightarrow$ 0.5. The score is the average over N samples:

$$S_{\text{Prompt}}(i) = \frac{1}{N} \sum_{n=1}^N x_i^n$$

The recommended models in the original paper are GPT-3 (text-davinci-003) or ChatGPT (gpt-3.5-turbo). However, considering the costs and convenience for further cross-lingual experiments, we use Qwen and Baichuan instead.

3 Experiments-Reproduction

3.1 Datasets

We used the primary dataset provided in the paper: The WikiBio GPT-3 Hallucination Dataset (238 human-annotated instances)[1] consists of GPT-3 generated biographies annotated at the sentence level for factual accuracy, with 20 sampled passages per instance for consistency checking. It includes 1,908 sentence-level factuality annotations (0.0 = Factual, 0.5 = Minor Hallucination, 1.0 = Major Hallucination).

3.2 Implementation

For our reproduction experiment, we used the original code provided in the paper for the four SelfCheckGPT methods excluding the Prompt method, with detailed adjustments made during the experimental testing process. As for the Prompt method where the model was replaced, we wrote the code for this method ourselves. We independently developed all the code for the testing part to evaluate the five methods.

First, we conducted the reproduction experiment on partial samples (50/238) to verify the correctness of the reproduced methods. Then we performed full-scale testing on all 238 articles in the dataset and compared the test results with the metrics reported in the original paper.

3.3 Evaluation

In order to compare with results in the original paper, we used the same evaluation metrics as the paper. There are three sentence-level evaluation metrics: NonFact/NonFact*/Factual AUC-PR. Due to the imbalanced dataset (more factual sentences), AUC-PR better reflects true model performance than AUC-ROC. And there are two passage-level metrics: Pearson correlation coefficient and Spearman correlation coefficient. They calculate the correlation between the model's passage-level scores and manual hallucination degrees to verify the model's transferability from sentence-level detection to passage-level evaluation.

3.4 Results

First, to save time and computational costs, we selected partial samples (50 out of 238 texts) for a preliminary test to verify the correctness of the reproduced methods. The results showed that the sentence-level metrics, namely NonFact and NonFact* AUC-PR, are both close to those reported in the original paper, but the Factual AUC-PR is generally lower. This indicated that our methods falsely classified some genuine factual sentences as "non-factual" or "hallucinatory," meaning the methods had a certain degree of false positive issue. Meanwhile, the passage-level metrics (Pearson and Spearman correlation coefficients) were generally lower compared to the results in the original paper.

We improved the code, which included adding a System Prompt with the correct dialogue template to SelfCheckPrompt to adapt to Chat models like Qwen-Chat, and other detailed adjustments. We selected the same partial samples (50 out of 238 texts) for retesting. As can be seen from the results, all metrics have been improved to a certain extent and are relatively close to those of the original paper. Although the Factual AUC-PR is still lower, and the Pearson and Spearman correlation coefficients are slightly inferior to the original results, this may be due to the small size of the test samples. Moreover, the metrics are generally within an acceptable range, which can basically confirm the correctness of the methods.

Table 1: Results of Testing on Partial Samples (50/238)

Method	Sentence-level(AUC-PR)			Passage-level(Corr.)	
	NonFact	NonFact*	Factual	Pearson	Spearman
Random	0.7296	0.2972	0.2704	-	-
SelfCheckGPT with BERTScore					
Reproducing Results	0.8250	0.5859	0.3543	0.5658	0.4677
Original Results	0.8196	0.4596	0.4423	0.5818	0.5590
SelfCheckGPT with NLI					
Reproducing Results	0.9222	0.5884	0.5888	0.7163	0.7116
Original Results	0.9250	0.4517	0.6608	0.7414	0.7378
SelfCheckGPT with MQAG					
Reproducing Results	0.7991	0.5194	0.3287	0.5423	0.5439
Original Results	0.8426	0.4006	0.4814	0.6107	0.5929
SelfCheckGPT with Ngram					
Reproducing Results	0.8835	0.4962	0.5440	0.6718	0.6431
Original Results	0.8563	0.4104	0.5847	0.6471	0.6491
SelfCheckGPT with Prompt					
Reproducing Results	0.9163	0.5978	0.5390	0.7353	0.6943
Original Results	0.9342	0.5319	0.6709	0.7832	0.7830

Then we conducted a full-scale experiment on all 238 texts, and the results are shown in the table below.

Table 2: Results of Full-Scale Testing on All 238 Articles

Method	Sentence-level(AUC-PR)			Passage-level(Corr.)	
	NonFact	NonFact*	Factual	Pearson	Spearman
Random	0.7296	0.2972	0.2704	-	-
SelfCheckGPT with BERTScore					
Reproducing Results	0.8120	0.5746	0.4347	0.5734	0.5441
Original Results	0.8196	0.4596	0.4423	0.5818	0.5590
SelfCheckGPT with NLI					
Reproducing Results	0.9250	0.5838	0.6608	0.7414	0.7378
Original Results	0.9250	0.4517	0.6608	0.7414	0.7378
SelfCheckGPT with MQAG					
Reproducing Results	0.8517	0.5074	0.5001	0.6487	0.6433
Original Results	0.8426	0.4006	0.4814	0.6107	0.5929
SelfCheckGPT with Ngram					
Reproducing Results	0.8562	0.5326	0.5843	0.6472	0.6491
Original Results	0.8563	0.4104	0.5847	0.6471	0.6491
SelfCheckGPT with Prompt					
Reproducing Results	0.9133	0.6190	0.5961	0.7196	0.7092
Original Results	0.9342	0.5319	0.6709	0.7832	0.7830

The results reveal that all metrics of the five methods are basically very close to those in the original paper, and some even exceed them. The methods with the best reproduction performance are MQAG and NLI: all five metrics of MQAG are higher than those in the original paper, while four metrics of NLI are exactly consistent with the original (and the remaining one is higher). The Prompt method has relatively poor reproduction

performance, as its Factual AUC-PR and passage-level metrics are lower than the original paper with a relatively significant gap. This may be attributed to the use of a different large language model in our work, resulting in performance discrepancies. In terms of the metric values themselves, although some metrics of the Prompt method are inferior to those in the original paper, they are still among the highest among the five methods. The best-performing methods among the reproduced ones are NLI and Prompt, which is also consistent with the findings of the original paper.

4 Experiments-Extension

Based on the reproduction experiments, we carried out the extension experiments. This part includes two extended research directions:

1. Ensemble method: Combine the top two methods via weighted averaging to verify performance improvement.

2. Cross-lingual evaluation: Use Chinese models on a Chinese dataset. Evaluate using sentence-level AUC-PR and passage-level correlation coefficients and compare Chinese/English models on Chinese data and English models on English data.

4.1 Ensemble experiments

Ensemble learning aims to enhance the performance of detection systems by leveraging the strengths of multiple base models. Given the superior performance of individual methods in the reproduction experiment, we designed a targeted ensemble experiment to explore whether fusing top-performing methods could break the performance ceiling of single models.

4.1.1 Datasets

The dataset used was consistent with the original paper and our reproduction experiment.

4.1.2 Evaluation

The evaluation metrics used were the same as those in the reproduction experiment, in order to align with the original paper's focus and ensure comparability across experiments.

4.1.3 Methods

1. Selection of Base Models

We selected SelfCheckNLI and SelfCheckPrompt as the base models for fusion, based on three core rationales:

1. Top-tier Performance: Since both methods ranked among the top two in both the original paper and our reproduction experiment, their consistent superiority confirmed them as the most reliable candidates for ensemble:

Original paper (NonFact AUC-PR): Prompt(0.9342) > NLI(0.9250) > Ngram(0.8563) > MQAG(0.8426) > BERTScore(0.8196).

Reproduction results (full-scale, NonFact AUC-PR): NLI(0.9250) > Prompt(0.9133) > Ngram(0.8562) > MQAG(0.8517) > BERTScore(0.8120).

2. Mechanistic Complementarity: The two methods differ fundamentally in architecture and working logic, minimizing overlapping errors. SelfCheckNLI relies on an encoder-only model (DeBERTa-v3-Large) fine-tuned on natural language inference (NLI) tasks, focusing on logical judgments of entailment or contradiction between sentences. SelfCheckPrompt uses a decoder-only generative model (Qwen-7B-Chat), emphasizing context-

aware reading comprehension and yes/no QA to assess semantic support.

3. Numerical Compatibility: Both methods output scores in the range [0.0, 1.0] (probability-like values), eliminating the need for complex normalization. This reduces noise introduced by preprocessing and simplifies the fusion process.

2. Ensemble Methods

We compared two weighting strategies to explore the optimal fusion scheme:

Scheme A: Simple Average: This method computes the unweighted average of scores from SelfCheckNLI and SelfCheckPrompt, with the formula:

$$\text{Ensemble Score}(i) = \frac{\text{NLI Score}(i) + \text{Prompt Score}(i)}{2}$$

Scheme B: Grid Search Weighting: This method optimizes the weight distribution between the two models via grid search to maximize performance on the full dataset. The formula is:

$$\text{Ensemble Score}(i) = \alpha \times \text{NLI Score}(i) + (1 - \alpha) \times \text{Prompt Score}(i)$$

Grid Search Range: α ranges from 0.0 to 1.0 with a step size of 0.05 (total 21 weight combinations).

It can find the theoretically optimal weight configuration for the specific dataset. Though it may have a risk of overfitting on small datasets, it's a standard practice in zero-resource hallucination detection research.

4.1.4 Results

1. Baseline Performance of Single Models

SelfCheckNLI: NonFact AUC-PR = 0.9250.

SelfCheckPrompt: NonFact AUC-PR = 0.9133.

2. Ensemble Performance

Both schemes outperformed the single-model baselines, with Scheme B achieving the best results:

Table 3: Results with Two Ensemble Methods

Ensemble Scheme	NonFact AUC-PR	Performance Gain (vs. Top Single Model)	Optimal Weight (NLI : Prompt)
A: Simple Average	0.9264	+0.0014(0.15%)	-
B: Grid Search Average	0.9299	+0.0049(0.53%)	0.8 : 0.2

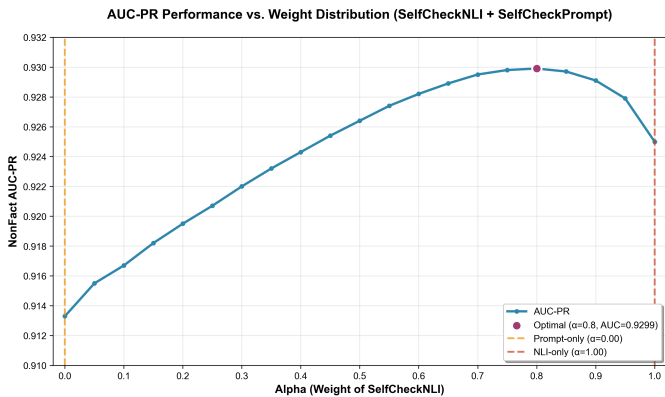


Figure 1: Grid Search Average Results with Different Weights

The grid search-based ensemble achieved the highest AUC-PR of 0.9299, representing a 0.53% improvement over the top single model (SelfCheckNLI, 0.9250). While the absolute gain seems modest, it is meaningful given the already high baseline

(AUC-PR > 0.92), where incremental improvements are typically challenging.

3. Analysis of Optimal Weight Distribution

The optimal weight configuration ($\alpha = 0.8$ for NLI, $1 - \alpha = 0.2$ for Prompt) reveals key insights into the roles of the two models:

SelfCheckNLI is the dominant model. Its higher weight (80%) reflects its superior robustness. As a model specialized in NLI tasks, DeBERTa-v3-Large outputs continuous, smooth, and logically consistent scores, making it the core of the ensemble.

SelfCheckPrompt performs as the "Corrector". Despite a lower weight (20%), it provides critical complementary value. The generative model's discrete outputs (0.0, 0.5, 1.0) lack granularity but excel in semantic understanding scenarios where NLI may misjudge (e.g., nuanced context-dependent factuality). It acts as a "second opinion" to correct partial false positives/negatives from NLI.

4.2 Cross-lingual experiments

The goal is to explore the generalization ability of SelfCheckGPT in low-resource, high-context languages (Chinese).

4.2.1 Datasets

To address the absence of dedicated Chinese hallucination detection datasets, we constructed the Chinese-WikiBio synthetic dataset by translating the original WikiBio dataset into Chinese.

4.2.2 Evaluation metrics

In this section, NonFact* was removed as Baichuan (serving as the judge) conducts binary classification (correct/incorrect) and cannot accurately distinguish minor from major errors; other metrics remain consistent with the reproduction experiment, including two sentence-level metrics (NonFact/Factual AUC-PR) and two passage-level metrics (Pearson/Spearman).

4.2.3 Baseline evaluation

Due to SelfCheckMQAG and SelfCheckNLI relying on English-specific models (T5-SQuAD, DeBERTa-MNLI) and thus being unavailable for Chinese, while SelfCheckBERTScore uses English RoBERTa by default, only SelfCheckNgram and SelfCheckPrompt were retained for cross-lingual transfer. Baichuan2-7B-Chat replaced the previously used Qwen for its strong performance in Chinese generation and instruction understanding. The results below shows that Prompt remains effective but Ngram failed.

Table 4: Baseline Evaluation of Cross-Lingual Testing

Language	Sentence-level(AUC-PR)		Passage-level(Corr.)	
	NonFact	Factual	Pearson	Spearman
SelfCheckGPT with n-gram				
Baichuan-Ngram (Chinese)	0.6090	0.3378	0.0463	0.0328
Reproduction (English)	0.8562	0.5843	0.6472	0.6491
SelfCheckGPT with Prompt				
Baichuan-Ngram (Chinese)	0.8069	0.6531	0.3522	0.3538
Reproduction (English)	0.9133	0.5961	0.7196	0.7092

4.2.4 Improved experiments

In order to to address the shortcomings of Chinese hallucination detection, we explored four improvement directions—

semantic enhancement, symbolic enhancement, reasoning enhancement, and translation transfer. While the results were not fully satisfactory, these attempts eliminated incorrect paths and uncovered underlying mechanisms.

1. Semantic Enhancement Path

As Ngram failed in the baseline evaluation, we attempted to enhance Ngram with RoBERTa embeddings first. Based on the original methods, we developed six variants (Semantic Ngram V1-V6) and an optimal transport (OT) based version, and each experiment built on the failures of the previous one. However, the performance was disappointing, with AUC scores even lower than the original N-gram baseline. The detailed results are listed below:

Table 5: Results of Semantic Enhancement Path

Experiment Method	Sentence-level metrics(AUC-PR)	
	NonFact	Factual
Reproduction(English Ngram)	0.8562	0.5843
Baseline(Chinese Ngram)	0.6090	0.3378
Semantic V1(Mean)	0.5585	0.3885
Semantic V2(Max-Min)	0.5878	0.4006
Semantic V3(Coverage)	0.5422	0.4424
Semantic V4(Sentence-Level)	0.5758	0.3986
Semantic V5(True BERTScore)	0.5520	0.3819
Semantic V6(IDF+Multi-Layer)	0.5473	0.3799
SelfCheck-OT(Whitened WMD)	0.5498	0.3812

(1) Semantic V1 (Mean): This variant introduced RoBERTa embeddings to address “synonym mismatch,” computing the average cosine similarity between tokens in the target sentence and sampled passages. The intent was to leverage embeddings’ ability to capture semantic similarity (e.g., “开心” vs. “快乐”) and thus improve N-gram performance. Steps included extracting character-level embeddings, computing token-wise cosine similarities, and using the average as the consistency score (lower score $\rightarrow$ higher hallucination probability). Result: NonFact AUC-PR = 0.5585 (worse than baseline). Failure was attributed to the averaging effect: hallucinatory sentences often contain only 1–2 erroneous tokens (e.g., wrong place names) while most tokens are factual; averaging masks the low similarity of those few erroneous tokens, making local hallucinations undetectable. Essentially, the high similarity of the background context overwhelms the negative signal from specific factual errors, leading to inflated consistency scores.

(2) Semantic V2 (Max-Min): To mitigate the “averaging trap,” we adopted a “short-board” strategy: a sentence is judged hallucinatory if any token exhibits low similarity. The average was replaced by the minimum similarity across all token pairs (lower minimum $\rightarrow$ higher hallucination probability). Result: NonFact AUC-PR = 0.5878 (slight improvement but still below baseline). Failure arose because RoBERTa’s embedding space shows anisotropy—even semantically related tokens can have low similarity due to noise. The min-similarity strategy became overly aggressive, misinterpreting normal semantic fluctuations as hallucinations and causing high false-positive rates.

(3) Semantic V3 (Coverage): Seeking a balance between the lenient mean (V1) and strict min (V2), we calculated a “pass rate”: set a similarity threshold (0.8), count the proportion of target tokens whose similarity to sampled tokens exceeds the threshold, and use this proportion as the consistency score (lower proportion $\rightarrow$ higher hallucination probability). Result: NonFact AUC-PR = 0.5422 (further decline). Failure occurred because embeddings were still computed at the character level. Chinese characters (e.g., “道” or “打”) are highly ambigu-

ous in isolation; without phrase context, character embeddings cannot accurately capture semantic similarity.

(4) Semantic V4 (Sentence-Level): To resolve “granularity misalignment,” we shifted from character-level to sentence-level embeddings: mean-pooling produced sentence embeddings for the target and each sampled passage, then averaged cosine similarities across passages. Result: NonFact AUC-PR = 0.5758 (no meaningful improvement). Failure resulted from sentence-level embeddings averaging out local semantic differences. For instance, the hallucinatory “李白出生在火星” and the factual “李白出生在碎叶城” share identical structure and high word overlap, leading to extremely high sentence-level similarity that submerges critical entity errors. This confirms that coarse-grained sentence representations lack the necessary resolution for fine-grained entity verification in hallucination detection tasks.

(5) Semantic V5 (True BERTScore) & V6 (IDF+Multi-Layer): V5 retained token sequences (“words in context”) to replicate BERTScore logic; V6 added IDF weighting and multi-layer fusion on top of V5. Steps included extracting contextualized tokens from RoBERTa’s last four layers, applying IDF weighting to emphasize low-frequency entity words, and computing IDF-weighted maximum cosine similarity. Results: NonFact AUC-PR(V5) = 0.5520, NonFact AUC-PR(V6) = 0.5473 (both too low). Failure attribution: The core task of hallucination detection is to distinguish “similar but distinct” entities (e.g., “长安” and “碎叶城”). Symbolic matching (Ngram) treats them as distant and easily detectable, but semantic embedding places them close due to shared category, making entity errors undetectable. Thus, for entity-level hallucinations, the fuzzy generalization of embeddings becomes a liability.

(6) OT: This approach applied manifold whitening to rectify vector anisotropy and used optimal transport (Wasserstein distance) to compute the minimal cost of matching sentence distributions, aiming to capture structural discrepancies for sensitive detection. Result: NonFact AUC-PR(OT) = 0.5498 (also failed).

2. Symbolic Enhancement Path

We tried symbolic enhancement path to improve Ngram in Chinese. We introduced Jieba part-of-speech tagging to filter for content words (KeyFact), retaining only nouns, verbs, and adjectives to reduce noise from function words. However, the result (NonFact AUC-PR = 0.5525) showed no meaningful improvement. This indicated that Jieba’s segmentation errors and the persistent synonym problem remained unresolved—simple function word filtering could not address the core issue of entity inconsistency.

3. Reasoning Enhancement Path

Although Prompt performed well, it still had potential to be improved. The standard SelfCheckPrompt method relies on direct yes/no questioning to assess the consistency between target sentences and sampled passages. This approach only invokes the model’s intuitive semantic judgment, ignoring its potential to analyze entities, relationships, and logical consistency explicitly. To explore whether enhancing reasoning could further improve detection accuracy, we designed two iterative Chain-of-Thought (CoT) prompting experiments (Zero-shot CoT & Few-shot CoT) aiming to guide the model to “think step-by-step” before concluding.

(1) Zero-shot CoT: No exemplars were provided; the model was instructed to reason freely. The prompt template was: “请分析句子中的每个实体和关系是否在上下文中一致。请逐步思考，列出一致或矛盾的点，最后得出结论。是否支持？(Analyze whether each entity and relationship in the sentence is consistent with the context. Think step-by-step, list points of

consistency or contradiction, and finally draw a conclusion. Is the sentence supported by the context? Answer Yes or No.)” Since the model outputs a long reasoning text, we extracted the final conclusion by searching for “Yes” or “No” in the last sentence (with ambiguous outputs labeled 0.5).

(2) Few-shot CoT (In-Context Learning): Zero-shot CoT might fail because the 7B model lacked guidance on reasoning structure. Few-shot CoT added 1 positive exemplar (factual sentence) and 1 negative exemplar (hallucinatory sentence) to the prompt, demonstrating the expected reasoning process. The template included exemplars with clear step-by-step reasoning (e.g., “Entity ‘Li Bai’ is mentioned in the context; relationship ‘born in Suyab’ is consistent → Conclusion: Yes”) followed by the target sentence and the same yes/no question.

Our experiments empirically falsify the hypothesis that CoT prompting outperforms direct QA for medium-scale LLMs in Chinese zero-resource hallucination detection. However, the results below show that both CoT variants underperformed the standard prompt significantly, with Few-shot CoT showing no improvement over Zero-shot CoT.

Table 6: Results of Reasoning Enhancement Path

Experiment Method	Sentence-level metrics(AUC-PR)	
	NonFact	Factual
Reproduction(English Prompt)	0.9133	0.5961
Baseline(Chinese Prompt)	0.8069	0.6531
CoT V1(Zero-Shot)	0.7059	0.6445
CoT V2(Few-Shot)	0.6996	0.5711

Three core mechanisms explain the failure: (1)CoT’s requirement for lengthy reasoning text triggers hallucination propagation. The model fabricates plausible but false justifications to “defend” erroneous entities, and these fabricated explanations mislead its final judgment. (2)There is a fundamental task-nature mismatch: hallucination detection in SelfCheckGPT relies on semantic perception (pattern matching between the target sentence and sampled passages) rather than deep logical reasoning. CoT forces the model to frame the task as syllogistic deduction, introducing unnecessary noise that undermines its optimized intuitive semantic judgment. (3)Few-shot CoT suffers from attention dilution. Lengthy exemplars in the prompt distract the 7B model’s limited attention, reducing its sensitivity to critical consistency signals and worsening performance compared to Zero-shot CoT.

Simple, direct yes/no prompts are optimal for medium-scale LLMs, as they avoid reasoning-induced noise and attention dispersion. While CoT fails to enhance performance, the analysis also hints that complementary strategies—leveraging the strengths of direct semantic perception (standard prompts) and mitigating blind spots via targeted optimizations—may offer paths to further improvement.

4. Translation Transfer Path

Since NLI performed best in the reproduction experiment but couldn’t be applied to Chinese, we translated Chinese texts back to English and used DeBERTa for cross-lingual NLI detection. This approach yielded an AUC of 0.6001, underperforming even the original Chinese Ngram method. Translation introduced secondary noise (Translation Noise), and the NLI model was insensitive to entity mismatches—often classifying such errors as “Neutral” instead of “Contradiction.”

4.2.5 Analysis

After many times of failure, and considering that we could get information of internal LLM output distributions (which was

different to the original paper), we took white-box metrics into account. We explore a deeper question: Do the external inconsistency scores (black-box metrics) align with the model’s internal uncertainty (white-box metrics)? In other words, can internal states linearly predict external hallucinations?

To address this, we defined two sets of metrics:

Black-box metrics included SelfCheck Prompt Score (measures inter-sample consistency, range: 0.0–1.0) and N-gram Score (statistical consistency based on lexical overlap), and white-box metrics included Predictive Entropy (quantifies token-level prediction uncertainty via $H(y|x) = -\sum p \log p$) and Perplexity (PPL, exponential form of entropy, reflecting overall fluency and confidence).

We designed an experiment to test the alignment between black-box and white-box metrics:

1. Data Generation: Use Baichuan2 to generate WikiBio biographies, retaining token-level Logits for white-box metric calculation.
2. Metric Computation: Calculate SelfCheck Prompt Scores (black-box) and average Entropy/PPL (white-box) for each sample.
3. Correlation Testing: Evaluate linear relationships via Pearson correlation coefficient and mean comparisons.

From the results we found that the Pearson correlation coefficient between white-box entropy and hallucination labels was ~ -0.0962 , indicating no meaningful linear relationship. In addition, Hallucinatory samples exhibited lower average entropy (2.336 vs. 2.365) and PPL (41.61 vs. 66.13) than factual samples. The model was more confident when generating false content than when stating facts.

Two core behavioral patterns explain the failure of white-box metrics: (1)Confident hallucinations. High-risk misattributions (e.g., assigning a celebrity’s biography to an obscure figure) exhibit extremely low PPL because the model recites high-frequency templates from training data. This fluency tricks white-box metrics into classifying false content as “absolute truth.” (2)Uncertain truths. Honest responses (e.g., refusing to fabricate information about unknown figures) are out-of-distribution(OOD) and trigger higher entropy/PPL due to the model’s hesitation, even though they are factually accurate.

In contrast, SelfCheckGPT’s black-box consistency check is irreplaceable: it exposes fluent hallucinations by leveraging the model’s inability to replicate the same false content across stochastic samples, regardless of how confident or fluent the generation is.

However, single-dimensional metrics (either black-box or white-box) have inherent blind spots: black-box metrics excel at identifying confident hallucinations but may miss OOD-related factual responses, while white-box metrics capture internal uncertainty tied to OOD generation. This suggests that by integrating black-box and white-box metrics to complement each other, and combining the external consistency of black-box methods and the internal uncertainty signals of white-box methods, we can mitigate individual limitations and further improve hallucination detection performance.

4.2.6 Final method

We proposed the Learning-to-Check (L2C) framework, which treats the outputs of multiple detection methods and model-internal metrics as features. Our objective was not merely to improve scores through machine learning-based fusion and achieve a higher NonFact AUC, but primarily to empirically test the necessity of external consistency when internal states are available.

1. Feature Set & Evaluation

We integrated black-box features (external consistency met-

rics) and white-box features (internal uncertainty metrics) to construct a multi-dimensional feature matrix. The features were the same as those discussed before. black-box features were Baichuan-Prompt and Baichuan-Ngram, and the white-box ones were token-level entropy and perplexity(PPL). All of them were normalized to the $[0, 1]$ range to ensure numerical compatibility.

We adopted 5-fold cross-validation to assess generalization, with sentence-level NonFact AUC-PR as the primary metric. The baseline model was Baichuan-Prompt, the best-performing single method in Chinese.

2. Fusion Methods

Firstly, we tried Logistic Regression (LR), which is a linear model that assumes additive relationships between features, optimizing weights to minimize classification loss.

LR’s gain was negligible and inconsistent, indicating limited ability to exploit feature complementarity. Since the linear model didn’t perform well, we tried to use a non-linear ensemble model. Random Forest(RF) is a non-linear ensemble model that uses multiple decision trees to capture complex, non-additive interactions between features, avoiding overfitting via bagging and feature randomness. The results are listed below:

Table 7: Results of LR & RF

Fold	NonFact AUC-PR (LR)		NonFact AUC-PR (RF)	
	Baseline(Prompt)	Fusion	Baseline(Prompt)	Fusion
1	0.7918	0.7912	0.7937	0.8041
2	0.7998	0.8062	0.8091	0.8230
3	0.7747	0.7741	0.7890	0.7937
4	0.8313	0.8348	0.8478	0.8528
5	0.8490	0.8492	0.8141	0.8296

We compared two representative models to explore the impact of linear vs. non-linear feature interactions:

Table 8: Comparison between Results of LR and RF

Model	Average NonFact AUC-PR	Average Baseline	Performance Gain vs. Baseline	Robustness (All 5 Folds)
Logistic Regression	0.8111	0.8093	+0.0018(0.22%)	3/5 folds with positive gain
Random Forest	0.8206	0.8107	+0.0099(1.22%)	5/5 folds with positive gain

Notably, the Random Forest achieved consistent performance improvements across all cross-validation fold, demonstrating strong generalization. The Random Forest performed better than the Logistic Regression, with an average performance gain of nearly 1.22%. The significant and robust gain of Random Forest validates the value of non-linear integration.

4.2.7 Analysis of Results

The non-linear model’s feature importance scores uncovered complex, context-dependent relationships, reviving previously underperforming features:

Table 9: Feature Importances (LR & RF)

Feature	Importance(LR)	Importance(RF)
Black-box Features		
Baichuan-Prompt	0.8549	0.5032
Baichuan-Ngram	-0.1691	0.1727
White-box Features		
PPL	-0.2080	0.1292
Entropy	0.0675	0.0892

The linear model(LR)’s feature coefficients reflected simplistic additive relationships, failing to leverage weak features effectively. The table shows that PPL had a negative weight, acting as noise due to linear misalignment with hallucination. The weight of Baichuan-Ngram was also negative, indicating statistical signals are distorted by Chinese word segmentation in linear settings.

For the non-linear model(RF), Baichuan-Prompt remained the core, but its dominance was reduced to accommodate complementary features. Baichuan-Ngram rose from negative weight to the second-most important, capturing non-linear statistical signals. The non-linear model revived weak features, as Ngram’s importance in Random Forest reveals its non-linear utility. In this way, when Prompt scores hover around 0.5 and the model becomes ambiguous, Ngram’s low scores serve as a “tiebreaker” to identify hallucinations. However, Linear models fail to capture this conditional relevance.

RF also leveraged white-box feature synergy, which was not made good use of when using linear models. Despite RLHF-induced overconfidence, residual uncertainty in deep probability distributions (PPL) is detectable via non-linear modeling, supplementing Prompt’s semantic judgments.

The test results showed consistent gains across 5 folds, confirming that the non-linear fusion strategy is not overfitted to specific data splits. It reliably extracts complementary information from multi-view features, mitigating the limitations of single models. The final method provides us with several key insights into Chinese hallucination detection:

1. Prompt is the core but not sufficient. It has blind spots that can be addressed by other features.
2. Chinese’s unique characteristics make statistical-semantic relationships inherently non-linear. Linear fusion cannot resolve this complexity, while tree-based models excel at it.
3. Multi-view fusion is the ultimate path to solve problems. When semantic understanding (Prompt) fails, statistical regularity (Ngram) and model uncertainty (PPL/Entropy) provide fallback signals.

5 Conclusion

This work completes a comprehensive reproduction, extension, and enhancement of SelfCheckGPT, delivering three key contributions. First, we successfully replicated the original framework, confirming the effectiveness of SelfCheckGPT. Second, the ensemble of NLI and Prompt provided a practical optimization for English hallucination detection. Third, our cross-lingual experiments uncovered unique challenges of Chinese hallucination detection, and the proposed L2C framework demonstrated that internal uncertainty metrics are insufficient due to model miscalibration. Even in grey-box scenarios where internal states are accessible, the semantic consistency captured by Self-CheckGPT remains the dominant hallucination indicator, while internal metrics serve only as auxiliary features used by non-linear models to correct edge cases. This highlights that while internal confidence reflects generation fluency, external sampling is indispensable for verifying factual correctness. This confirmed that multi-view feature integration—rather than single-dimensional improvements—is the optimal path for Chinese hallucination detection.

Overall, this study provides insights into consistency-based hallucination detection and its cross-lingual adaptation, offering practical solutions for reliable LLM output verification.

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